

GROUP PROJECT

December 10

Course

**Digital
Logic
Design**

SCENARIO

Scenario 2 Access Control & Indicators



Scenario 3 Mini Vending Controller



MINI VENDING CONTROLLER

Inputs

- C1 (1-unit coin)
- C2 (2-unit coin)
- S (Select)
- O (Out of stock)

Outputs

- Dispense
- ReturnCoin
- ThankLED
- Change
- Error
- ServiceLED

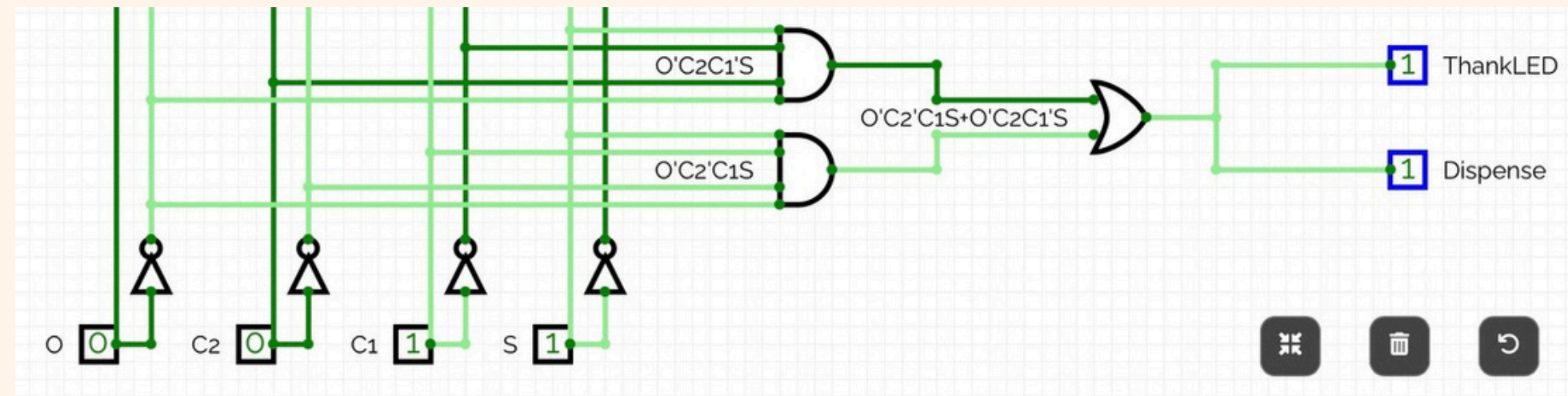
MINI VENDING CONTROLLER

- Dispense

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (3,5)$

$$SOP = O' C2' C1 S + O' C2 C1' S$$

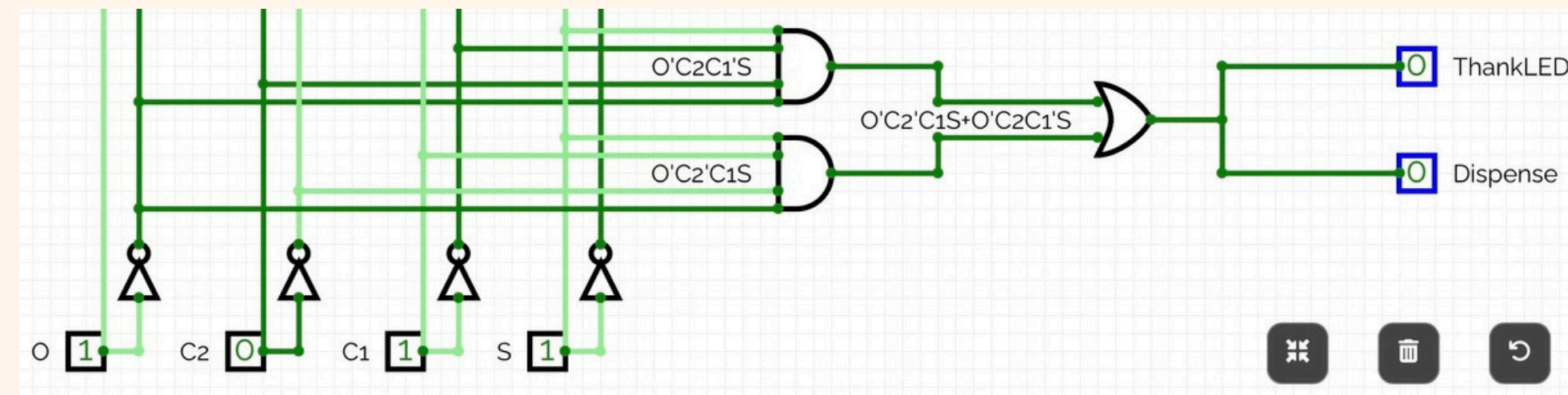


- ThankLED :

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (3,5)$

$$SOP = O' C2' C1 S + O' C2 C1' S$$



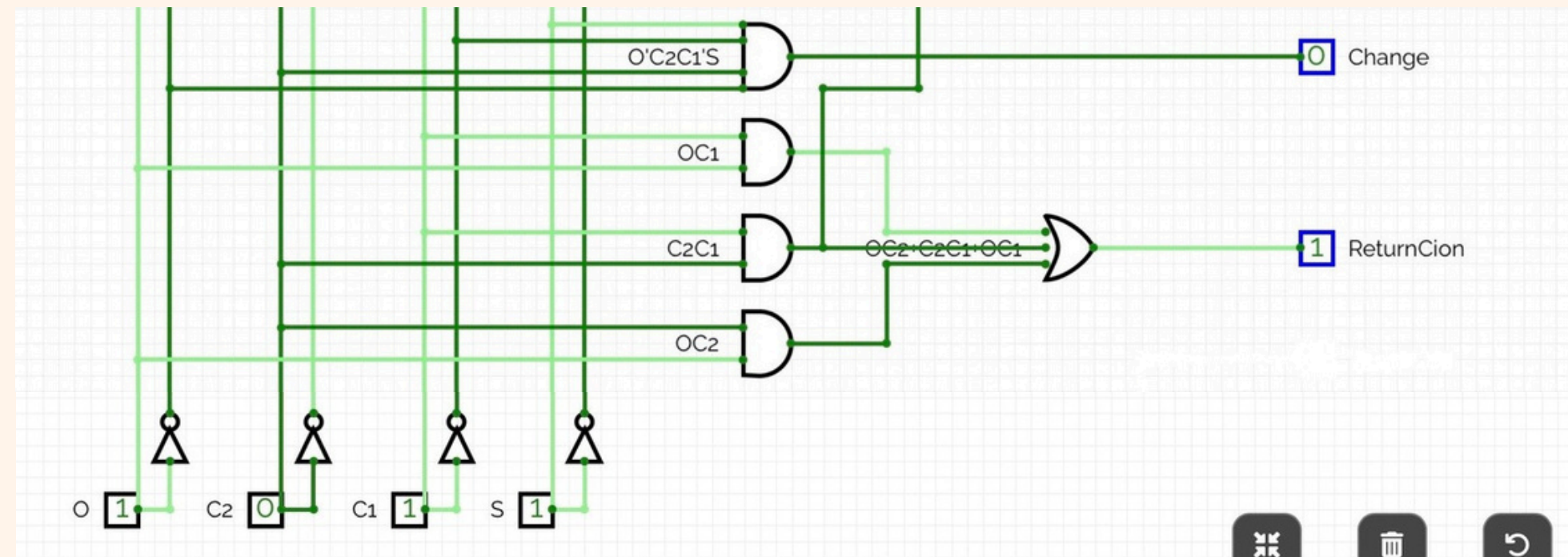
MINI VENDING CONTROLLER

- ReturnCoin

0	0	0	0
0	1	1	1
1	1	1	1
0	0	1	1

Minterms: $\Sigma (6,7,10,11,12,13,14,15)$

$$\text{SOP} = O' C2 + C2 C1 + O C2$$

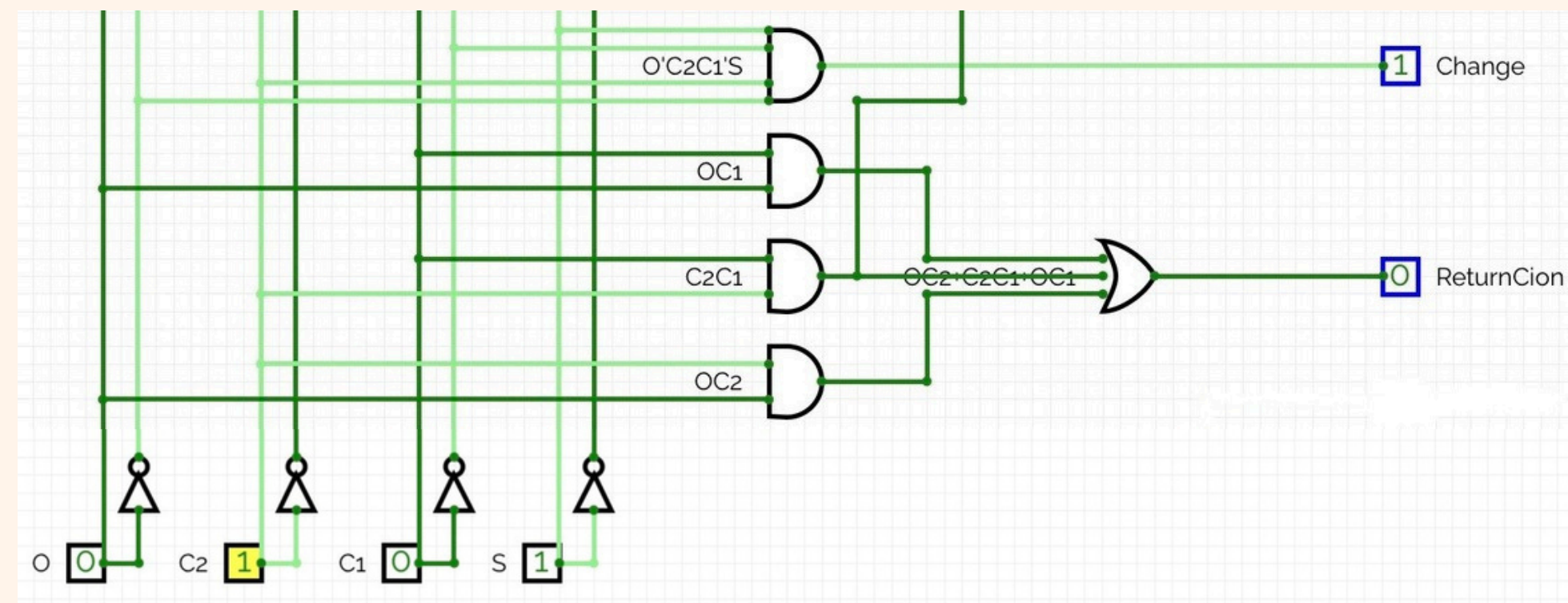


- Change :

0	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (5)$

$$\text{SOP} = O' C2 C1'$$



MINI VENDING CONTROLLER

- Error:

0	1	1	0
0	1	1	1
1	1	1	1
1	1	1	1

Minterms: $\Sigma (1,6,7,8,9,10,11,12,13,14,15)$

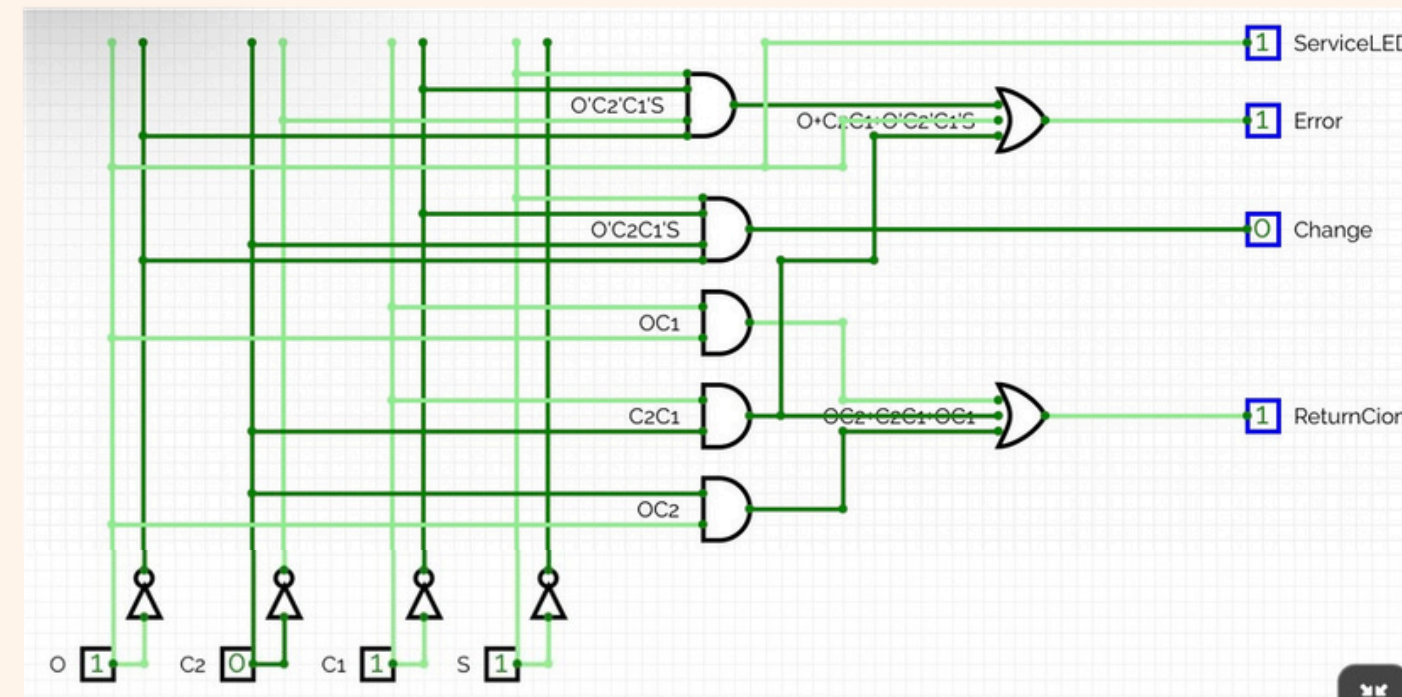
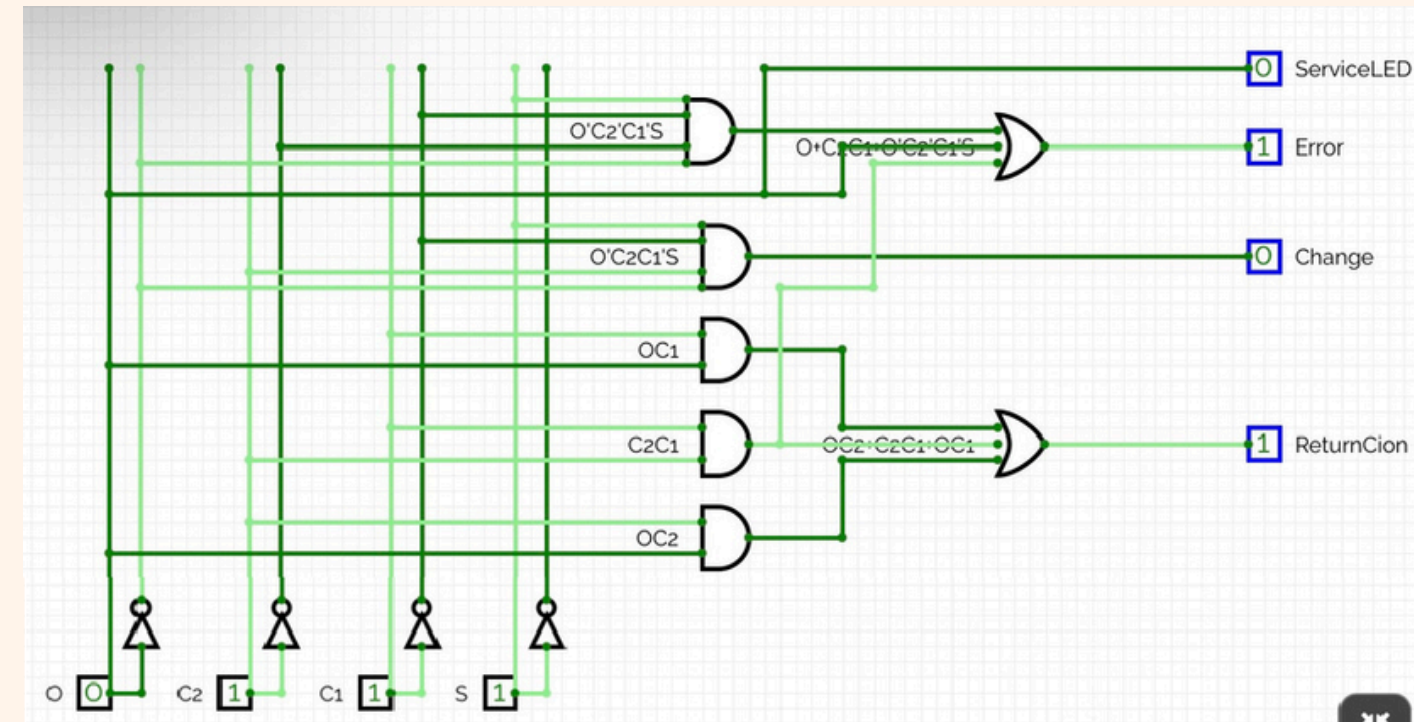
SOP = $O + C_2 C_1 + O' C_2' C_1' S$

- ServiceLED :

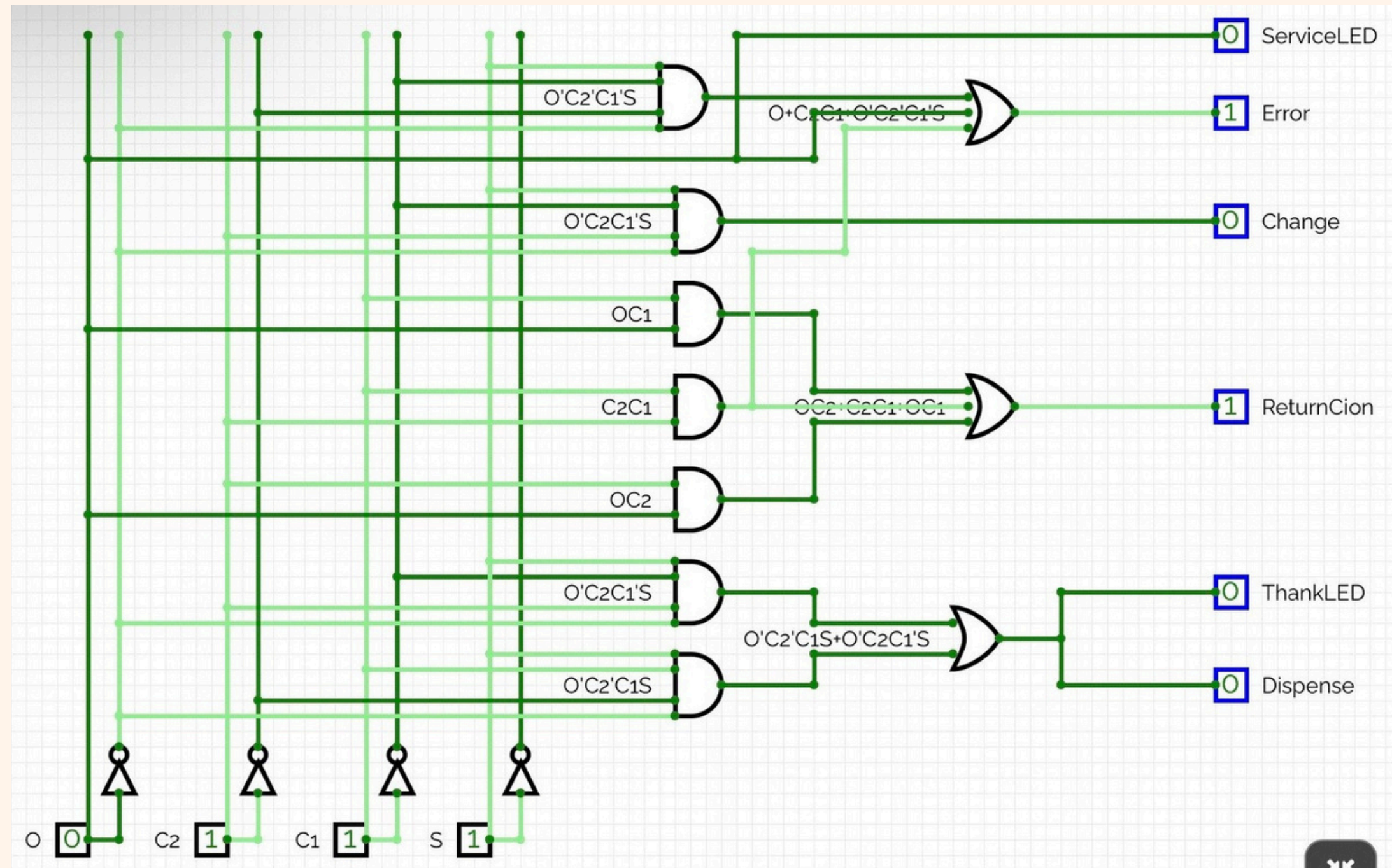
0	0	0	0
0	1	0	0
1	1	1	1
1	1	1	1

Minterms: $\Sigma (8,9,10,11,12,13,14,15)$

SOP = 0



MINI VENDING CONTROLLER



ACCESS CONTROL & INDICATORS

Inputs

- A (Admin)
- G (Guest)
- D (Door)
- F (Fire)

Outputs

- Grant
- Deny
- RedLED
- GreenLED
- Lock
- Evacuate

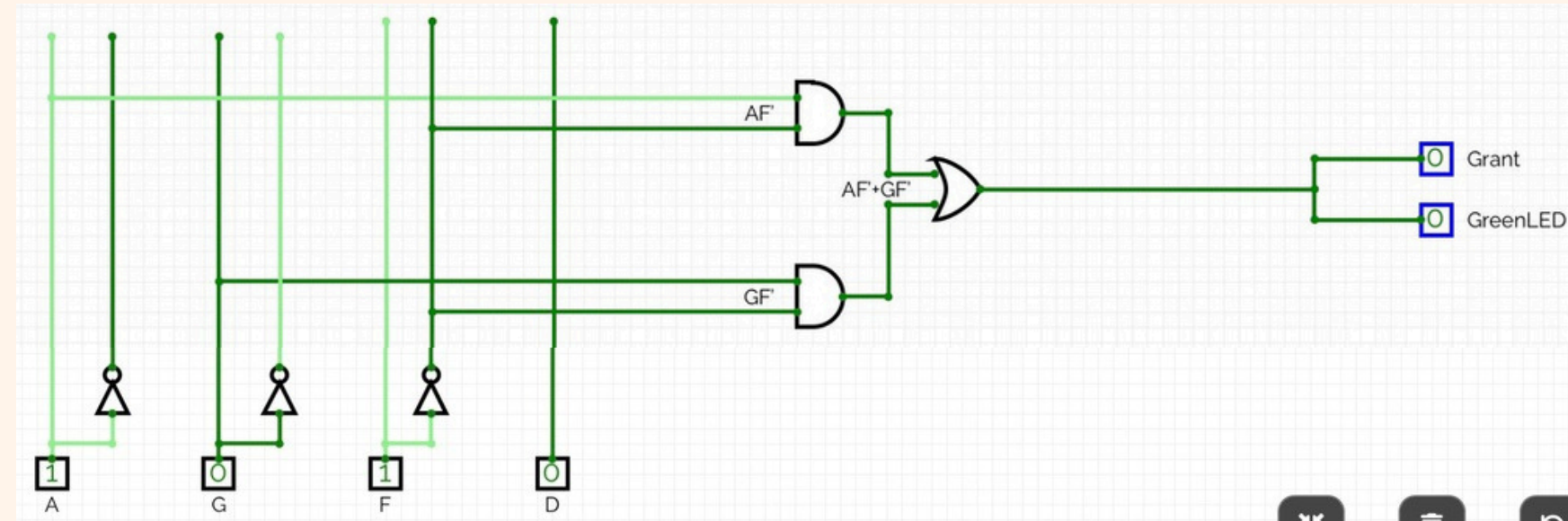
ACCESS CONTROL & INDICATORS

- Grant

0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

Minterms: $\Sigma (4, 6, 8, 10, 12, 14)$

$$SOP = A F' + G F'$$

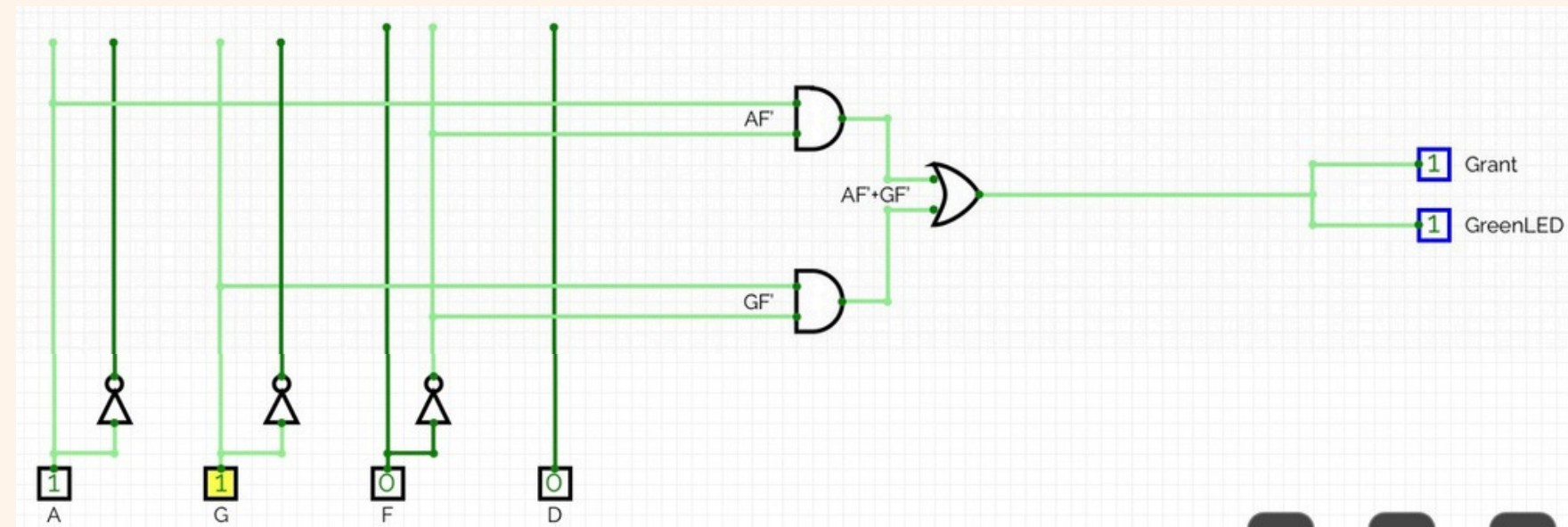


- GreenLED :

0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

Minterms: $\Sigma (4, 6, 8, 10, 12, 14)$

$$SOP = A F' + G F'$$



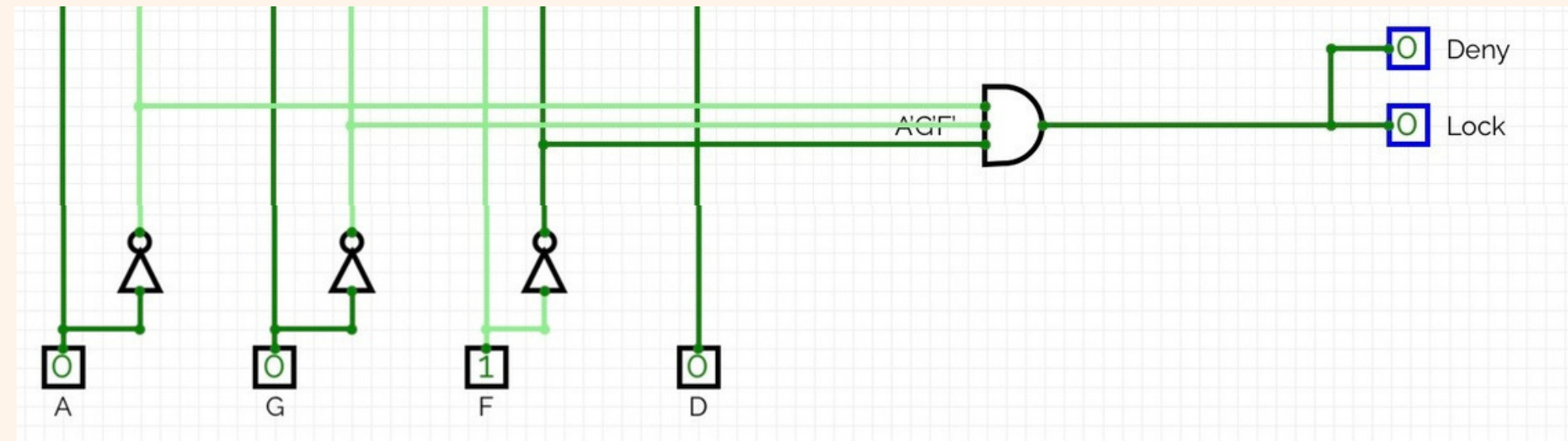
ACCESS CONTROL & INDICATORS

- Lock :

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (0, 2)$

SOP = $A' G' F'$

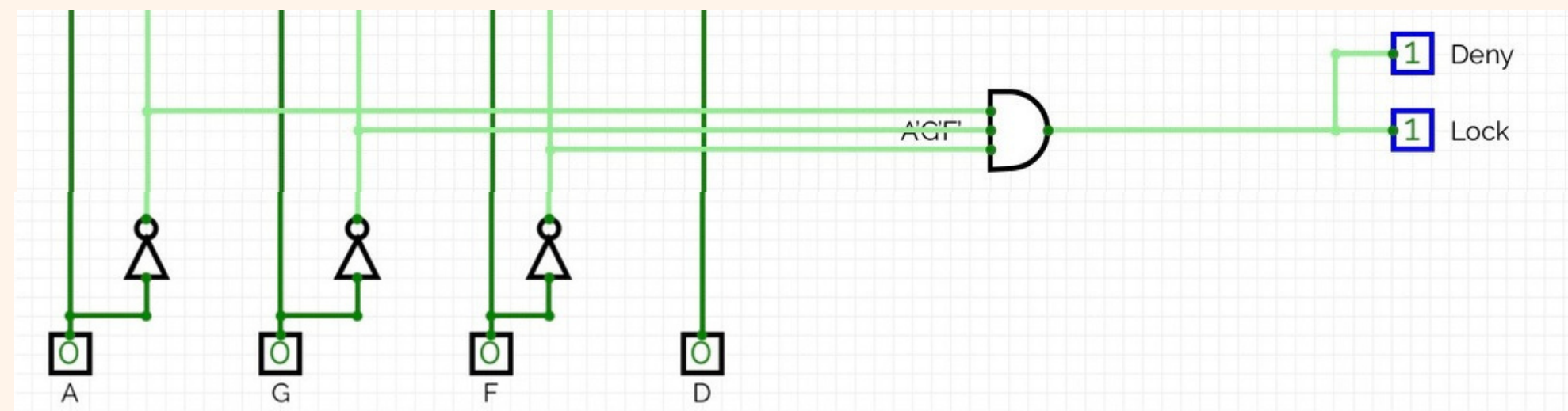


- Deny

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (0, 2)$

SOP = $A' G' F$



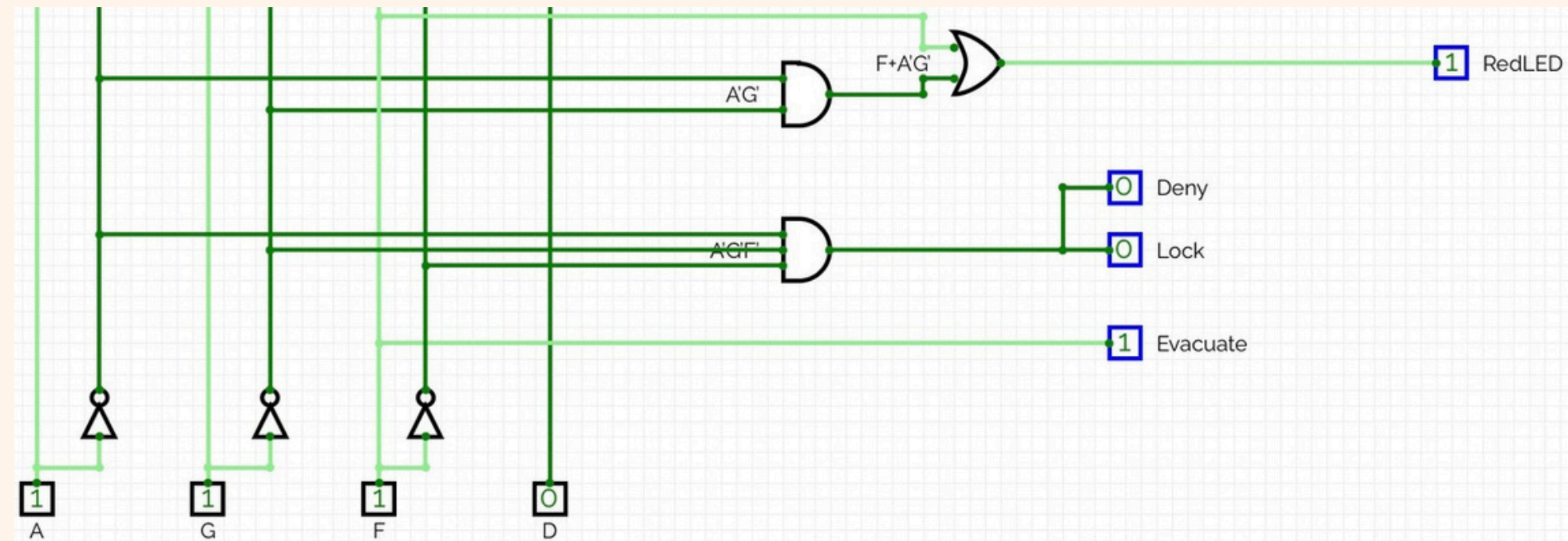
ACCESS CONTROL & INDICATORS

- RedLED :

1	1	1	1
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: $\Sigma (0, 1, 2, 3, 5, 7, 9, 11, 13, 15)$

$$\text{SOP} = F + A' G'$$

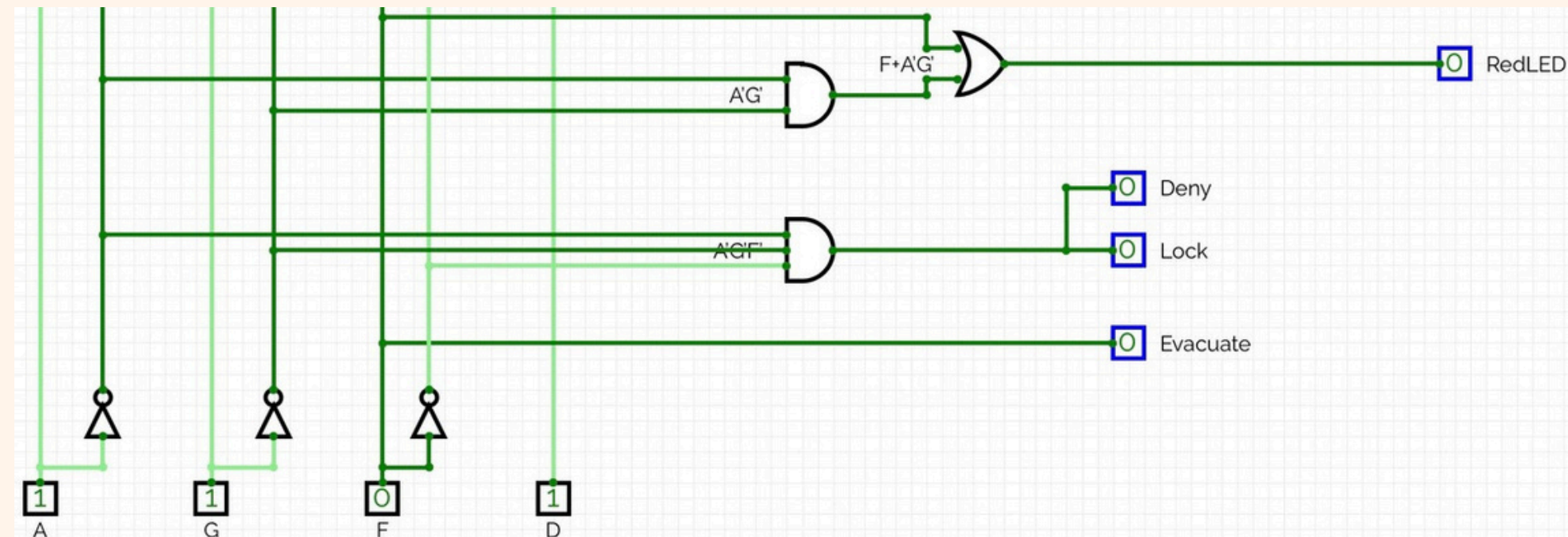


- Evacuate :

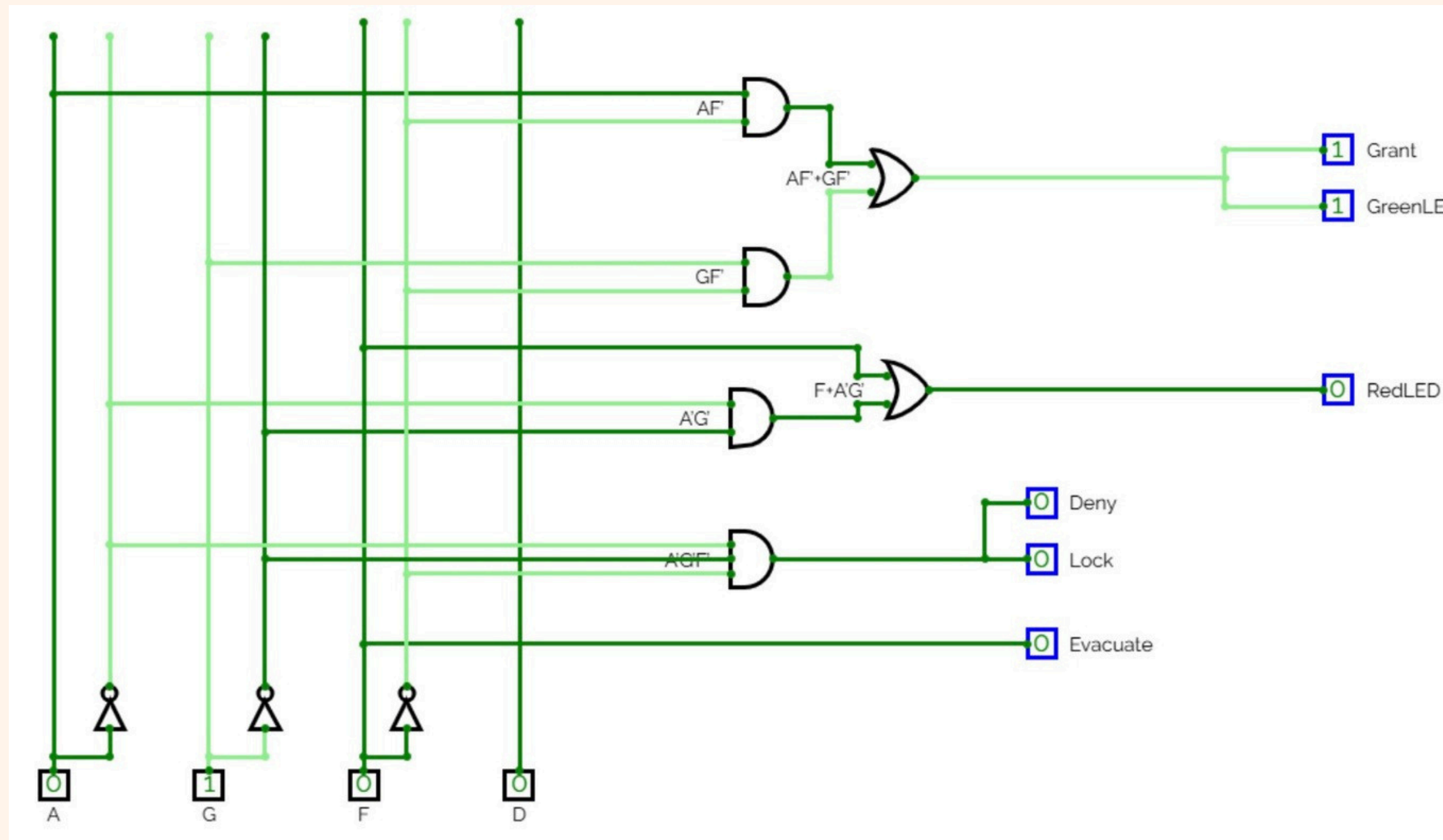
0	1	1	0
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: $\Sigma (1, 3, 5, 7, 9, 11, 13, 15)$

$$\text{SOP} = F$$



ACCESS CONTROL & INDICATORS



USAGE SUGGESTIONS

University gate

Machines in streets

A grayscale photograph of a mountain range. The foreground shows dark, forested slopes. In the background, several mountain peaks are visible, with the most prominent one on the right. The sky is filled with soft, wispy clouds. The text 'THANK YOU' is centered in the upper half of the image in a large, bold, black sans-serif font.

**THANK
YOU**